

Continued from last class $L: P_2 \rightarrow P_1$

L "in coordinates"

$$L(x^2) = 0 \quad L(x) = x \quad L(1) = 1$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} L(ax^2 + bx + c) &= aL(x)^2 + bL(x) + cL(1) \\ &= a(0) + b(x) + c(1) \\ &= bx + c \end{aligned}$$

What does this computation look like in coordinates

$(2 \times 3)(3 \times 1) = 2 \times 1$ matrix multiplication

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ c \end{bmatrix} = a \begin{bmatrix} 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0(a) + 1(b) + 0(c) \\ 0(a) + 0(b) + 1(c) \end{bmatrix}$$

If $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and we choose a basis for the domain and codomain $[b_1, b_2, b_3]$

$$A(b_1) = a_{11}b_1 + a_{12}b_2 + a_{13}b_3$$

$$A(b_2) = a_{21}b_1 + a_{22}b_2 + a_{23}b_3$$

$$A(b_3) = a_{31}b_1 + a_{32}b_2 + a_{33}b_3$$

then the "multiplication"

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{Compute } A(xb_1 + yb_2 + zb_3) \text{ in coordinates}$$

$$= \begin{bmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \\ a_{31}x + a_{32}y + a_{33}z \end{bmatrix}$$

$(3 \times 3)(3 \times 1) = (3 \times 1)$
 for $m \times n$ matrix
 # of rows $= m$ is the dim of the codomain of A , n is the dim of the domain of A

Ex: $L: P_2 \rightarrow \mathbb{R}$ where $L(x^2+x) = 0$, $L(x+1) = 1$, $L(x-1) = -1$

$$[x^2, x, 1]$$

our bases

$$\downarrow$$

$$[1]$$

$$L(x^2) = ?$$

$$x^2 = a(x^2+x) + b(x+1) + c(x-1)$$

$$x^2 = a(x^2) + (a+b+c)x + (b-c)$$

$$x^2 = ax^2 + (a+2b)x$$

$$b=c$$

$$\downarrow (a+2b)x = 0$$

$$a = -2b$$

$$\frac{1}{2} = b = c$$

$$L(x^2) = L(x^2+x) - \frac{1}{2}L(x+1) - \frac{1}{2}L(x-1) = 0$$

$$x = a(x^2+x) + b(x+1) + c(x-1)$$

$$a=0$$

$$x = (b+c)x + (b-c)$$

$$x = 2bx$$

$$b = \frac{1}{2}$$

$$b=c$$

$$c = \frac{1}{2}$$

$$L(x) = 0 \quad L(x+1) = 1$$

$$L(1) = 1$$

$$L(x) = \frac{1}{2}L(x+1) + \frac{1}{2}L(x-1)$$

$$= 0$$

$$L(ax^2+bx+c)$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} c \end{bmatrix}$$